



Weichi Zhao

AI Infrastructure / LLM Serving Platform Engineer

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Target roles: LLM serving platform, OpenAI-compatible MaaS, AI gateway, agent platform, heterogeneous GPU/XPU scheduling.

Profile

AI infrastructure engineer focused on production LLM serving platforms and model-service control planes. I build OpenAI-compatible MaaS capabilities, offline batch inference, model gateways, routing/orchestration reliability, and Kubernetes-native serving on heterogeneous GPU/XPU clusters. My strength is turning research and model workloads into reusable platform primitives with observable routing, online/offline traffic isolation, rollout safety, and incident-driven hardening.

Impact Highlights

- Scaled BatchFlow for production data distillation: over 2.76M requests / 23.9B tokens in a holiday run; later around 10M requests per day with success rate above 99%.
- Validated BatchFlow -> SGLang Router -> multi-KLX backend for Qwen3: about 300K requests / 9.75B tokens, about 99.9% success rate, and below 5% BatchFlow/Router CPU and memory overhead.
- Hardened serving reliability across gateway, Router, Worker, Service discovery, readiness, and long-connection failure modes instead of relying only on Pod-level health.

Core Skills

LLM Serving

SGLang, vLLM, OpenAI-compatible APIs, streaming, embeddings, Responses API, PD disaggregation, KV/prefix cache, cache-aware and bucket routing.

MaaS Gateway

Higress, LiteLLM, AI Gateway, authentication, model mapping, token-level rate limiting, route synchronization, observability.

Control Plane

Kubernetes operators/controllers, CRDs, Helm, Kustomize, readiness, finalizers, service discovery, rollout safety, retry and rollback workflows.

Agent Platform

Private-agent protocol integration, tool-calling API compatibility, Claude/OpenAI-style adapters, workflow safety, local agent governance.

Engineering

Go, Python, Java, Linux, Prometheus metrics, MySQL/SQLite/Redis, distributed systems debugging, performance and capacity testing.

Experience

Zhejiang Lab, High-Performance Computing Facility Research Center

2024 - Present

Senior Research Specialist, AI infrastructure and MaaS platform

- Built and operated LLM serving platform capabilities across CPU ACK and GPU ACK clusters, covering private model deployments through Higress, public model services through LiteLLM, SGLang Router based model routing, IGD/IGDSA inference orchestration, BatchFlow offline inference, and Agent-facing API compatibility.
- Owned production troubleshooting and hardening for LLM serving paths, including Router worker registration convergence, Service/Endpoint readiness, PD prefill/decode topology, CrashLoop and long-connection failures, gateway header/model mapping issues, and multi-cluster rollout validation.
- Designed MaaS-Controller and optimizer-loop mechanisms around LiteLLM, Router, IGDSA, BatchFlow, and gateway metrics, with public-model limit adjustment, offline concurrency tuning, observation windows, rollback criteria, and online-SLA-first rules.

Zhejiang Lab, Intelligent Supercomputing Research Center

Aug 2021 - 2023

Engineering Specialist, cloud-native AI and HPC platforms

- Worked on cloud-native AI/HPC platform projects, including Kubernetes adaptation on domestic OS and accelerators, device plugin customization for accelerator availability, PyTorch/model adaptation, and distributed training framework exploration.
- Supported model training and inference platform adaptation in domestic heterogeneous hardware/software environments, covering job submission, resource discovery, container runtime validation, and baseline performance verification.

Selected AI Infrastructure Projects

BatchFlow - OpenAI-compatible offline inference platform

LLM batch inference

Asynchronous JSONL batch execution for large-scale, non-real-time model workloads.

- Designed Batch API compatible capabilities: file upload, batch lifecycle, cancellation, result/error files, streaming file processing, request-level error isolation, and backend concurrency control.
- Refactored BatchFlow toward multi-replica operation with Kubernetes leader election, sharded workers, header forwarding, upload limits, stale-file cleanup, Kustomize upgrades, monitoring, and failure-tolerant finalization.
- Supported production data-distillation workloads: over 2.76M requests and about 23.9B tokens processed during a holiday run; later supported daily processing around 10M requests with success rate above 99% under controlled concurrency.
- Validated a BatchFlow -> SGLang Router -> multi-KLX path for Qwen3-235B-A22B-Thinking-2507 at about 300K requests, about 9.75B tokens, about 99.9% success rate, and below 5% BatchFlow/Router CPU/memory overhead.

SGLang Router, IGD, and production serving reliability

Routing and orchestration

Reliable worker registration, routing, and recovery for multi-worker and PD-disaggregated inference.

- Improved Router worker synchronization by reconciling desired workers against live `/workers` state and requeueing on missing, extra, or failed registrations.
- Enhanced IGD/ISI startup and readiness with Pod-IP early registration, `publishNotReadyAddresses`, preserved Service finalizers, Router metrics Service generation, and ISI-level auto-restart with retry budgets.
- Led production RCA for ARG_MAX overflow from service-link environment variables, Router not managed after Worker Ready, stale Service selectors, and long-connection timeout behavior.

MaaS gateway, control plane, and agent-compatible access

Higress, LiteLLM, Agent APIs

Unified model access, routing, rate limiting, protocol compatibility, and observability for private and public model services.

- Participated in the platform architecture that separates Higress private entry, LiteLLM public entry, SGLang Router routing, BatchFlow offline execution, and MaaS-Controller coordination.
- Validated AI gateway capabilities including token-level rate limits, model mapping, route synchronization, metrics collection, OpenAI-compatible protocols, embeddings, and Responses-style API boundaries.
- Mapped private and public model entries to OpenAI-compatible surfaces used by agent clients, covering Chat Completions, Responses-style APIs, embeddings, and tool-calling workflows.
- Resolved user-facing integration issues by tracing requests across gateway, router, worker, service discovery, and model protocol layers instead of relying only on top-level deployment health.

Kubernetes device plugin and heterogeneous accelerator platform

GPU/XPU scheduling

Cloud-native resource discovery, scheduling, and validation for domestic heterogeneous accelerator environments.

- Customized device-plugin and resource-management paths for accelerator availability, node resource reporting, container runtime validation, card-level scheduling, and failure isolation.
- Built deployment, upgrade, rollback, and E2E validation procedures for Kubernetes AI/HPC environments on domestic operating systems and accelerator stacks.

Earlier Projects

Arizona State University, SNAC Lab

Jan 2020 - Aug 2020

Virtual platform developer and research assistant

- Developed modules for THOTH Lab, an open virtual lab platform for education and research, including SDN/OpenFlow lab modules and AWS instance control through Boto3-backed REST APIs.
- Implemented an Android prototype for an IR-CP-ABE based network security research system using Java.

Arizona State University, CIDSE

Aug 2019 - Dec 2020

Graduate service assistant

- Assisted CSE205 Object-Oriented Programming, CSE412 Database Management, CSE434 Computer Networks, and CSE464 Software QA and Testing.

Education

Arizona State University

May 2019 - Dec 2020

Master of Computer Science, GPA 3.97/4.00

Arizona State University

Aug 2015 - May 2019

Bachelor of Science in Computer Science, GPA 3.40/4.00